Caroline Segestaal

Profile

Newly graduated with an MSc in Artificial Intelligence for Medicine and Medical Research from University College Dublin. Experienced in applying machine learning and data analysis techniques to real-world clinical datasets, with a strong interest in biomedical research, healthcare AI and translational data science. Skilled in developing reproducible machine learning pipelines using Python and modern ML frameworks. My work combines ML with statistical methods to analyse clinical and biomedical data to understand diseaeses.

Education

University College Dublin, MSc in AI for Medicine and Medical Research (2025 - 2026)

Master of Science, First Class

- Developing AI and machine learning solutions for biomedical and clinical research through practical projects.
- During an internship with a research group I developed an end-to-end machine learning pipeline for predicting pancreatic cancer from metabolomics data. Final model achieved a PRAUC of 97.4% and an accuracy of 91.3%.
- Built a 3D segmentation residual network with deep supervision for the segmentation of foetal brain MRIs, which achieved a mean accuracy of 82% across reconstruction and tissue types.
- Studying advanced machine learning, medical AI, statistical methods, biostatistics, biomedical data analysis and research methodologies.

Freestanding Master Courses (2024 - 2025)

- Applied Pharmaceutical Bioinformatics, Uppsala University, 5 ECTS.
- Pharmaceutical Bioinformatics, Uppsala University, 7.5 ECTS.
- Smart Healthcare with Applications, Halmstad University, 4 ECTS.

Swansea University, BSc Computer Science (2021 - 2024)

First Class Honours

- Awarded 'Best Computer Science project of year 2024',
- Deepened knowledge of algorithms, programming languages and software engineering principles, with extra focus on logic, big data and machine learning.

Dissertation: *Machine Learning Assessment of Novel Biomarkers for Aggressive Prostate Cancer:*

- Exploring multiple ML models to assess novel biomarkers in their potential usability for diagnosing prostate cancer as an alternative to prostate biopsy, using real medical data from Wales Cancer Biobank.
- Achieved overall accuracies of up to 88% using XGBoost.
- Achieved a 95% grade on the dissertation.

Technical Experience

Machine Learning & Clinical Data Analysis

- Developed end-to-end machine learning pipelines using Python, Pandas, NumPy, Scikit-learn, TensorFlow and PyTorch.
- Performed preprocessing, validation and exploratory analysis of clinical datasets.
- Produced data visualisations using Matplotlib and Seaborn to communicate analytical findings.
- Projects involved building survival, disease prediction/prognosis, ODE and medicine response models using real medical data in both R and Python.

Software Engineering Practices

- Version control using Git and GitHub,
- Writing clean, maintainable, well-documented, and well-structured code,
- Familiarity with data pipelines and workflow structuring.

Technical Skills
Programming: Python, R, Java, SQL
Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn
Tools: Git, GitHub
Research: Clinical data analysis, machine learning pipelines, model evaluation, data preprocessing, data cleaning
Languages: English + Swedish (fluent), Spanish (B1.2)

Work Experience

Content Moderator, Majorel, Barcelona (2020 - 2021)

- Developed strong pattern recognition and anomaly detection skills by reviewing 30-60 live streams simultaneously and correctly applying the platform's rules with 98% accuracy.

Customer Service Agent, Webhelp, Barcelona (2020)

- Worked with internal systems to resolve logistics and data-related issues,
- Collaborated with cross-functional teams, operations, warehousing, support, in multiple languages (English, Swedish, Spanish)
- Maintained accurate records in fast-paced, data-driven environment.

Nespresso Coffee Advisor, Sweden (2018-2019, 2021)

- Trained new staff and communicated complex product information clearly,
- Built strong customer communication and teamwork skills.